# Distinctive Features

Dr. Josef Fruehwald

## Phonemes - Contrast

# p ≠ b

- pin
- nap
- tripe
- bin
- nab
- tribe

## Phonemes - Contrast

# l ≠ p

- lick
- pipe
- pine
- pick
- pile
- line

## Phonemes Contrast

# t ≠ d

- time
- bit
- tome
- dime
- bid

- dome

## Phonemes - Contrast

Is the *way* that

# p ≠ b

more similar to how

# l ≠ p

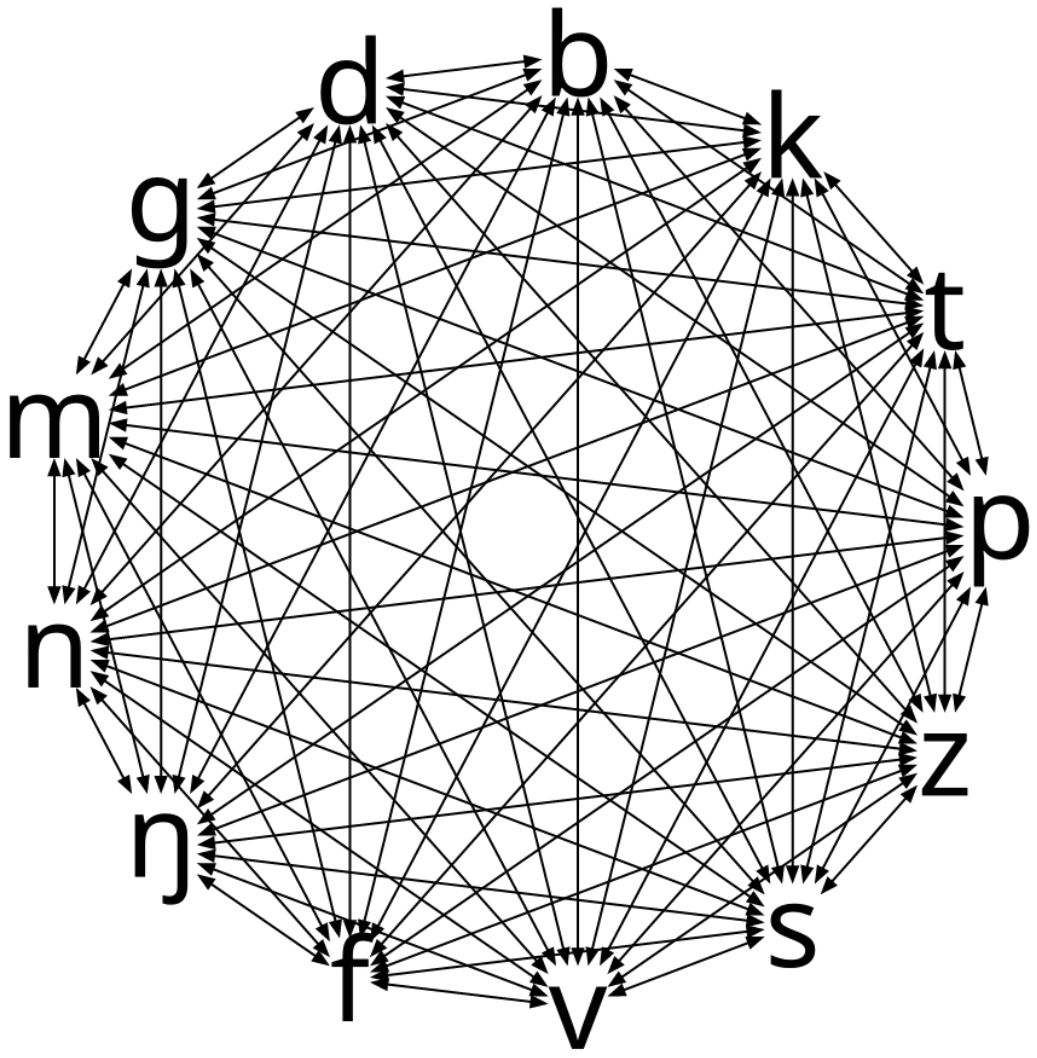
or to to how

# t ≠ d

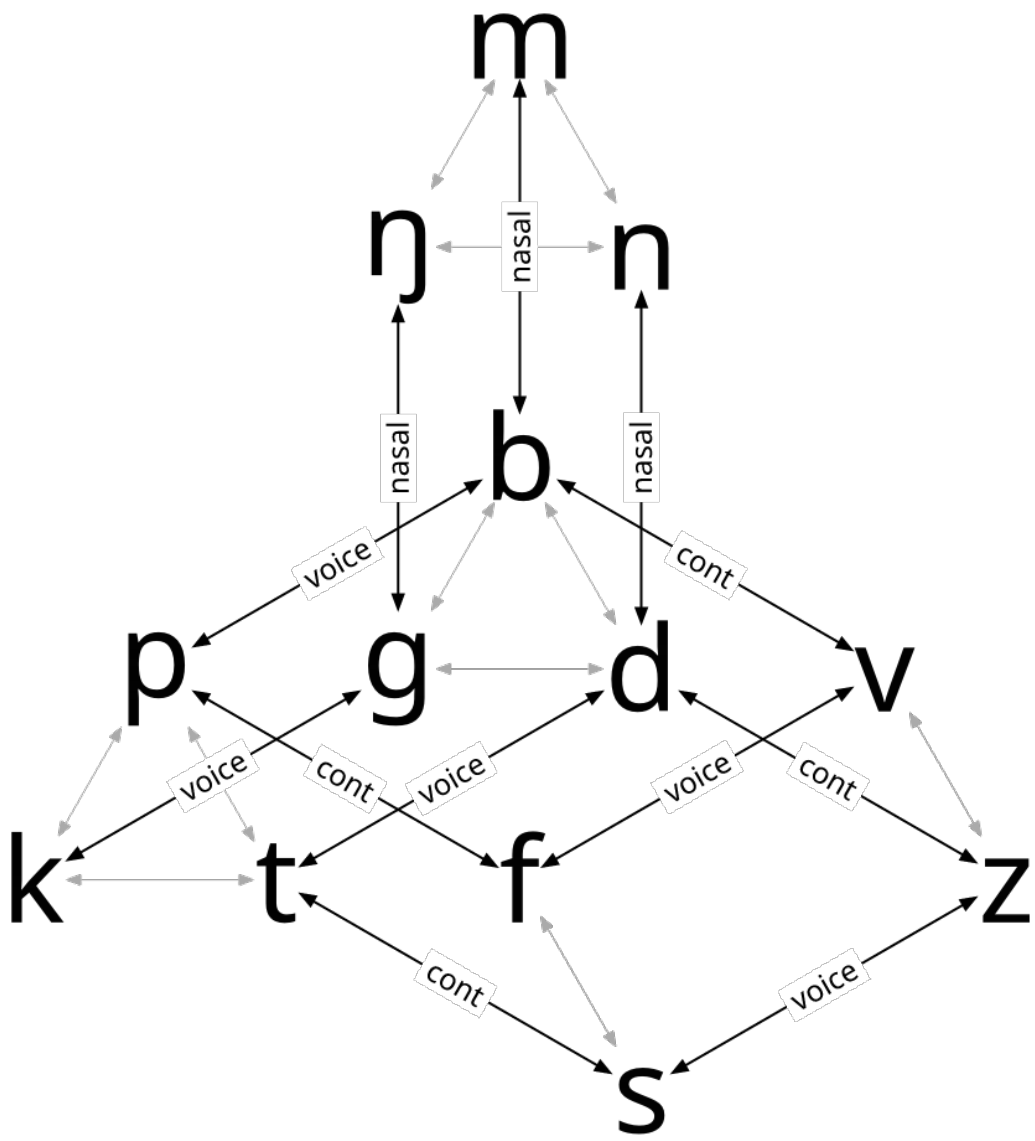
?

## Phonemes - Contrast

If all contrasts are the same, then...

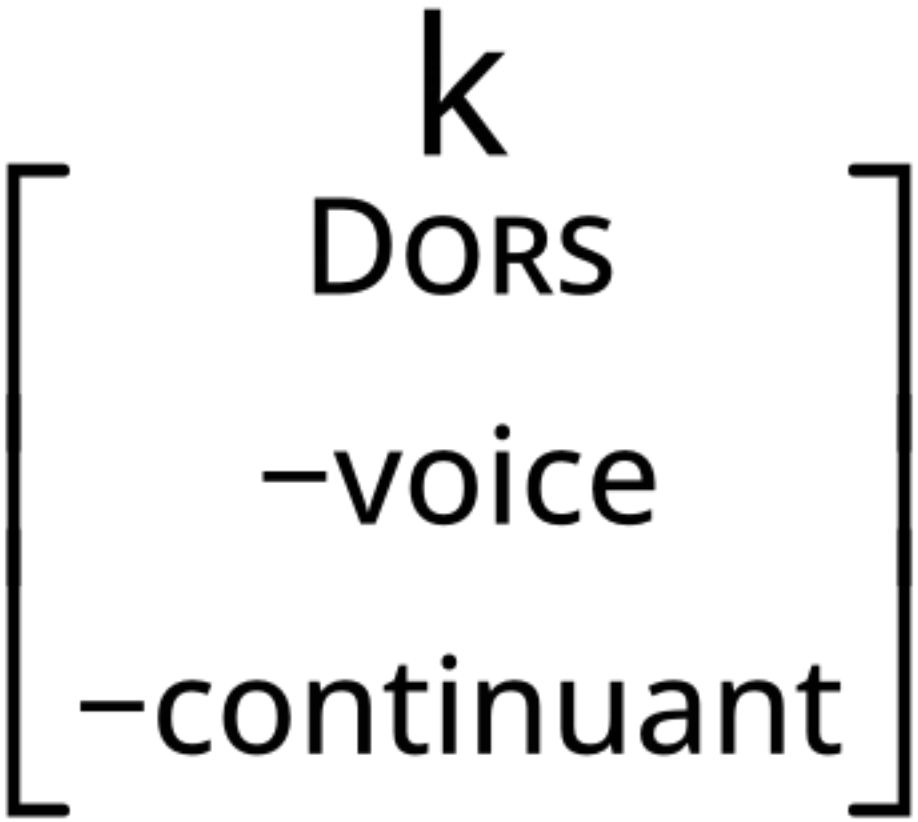

b
k
t
p
z
s
v
f
ŋ
n
m
g
d

## Distinctive Features




## Distinctive Features

Each phoneme is represented as the set of distinctive features that distinguishes it from all other phonemes



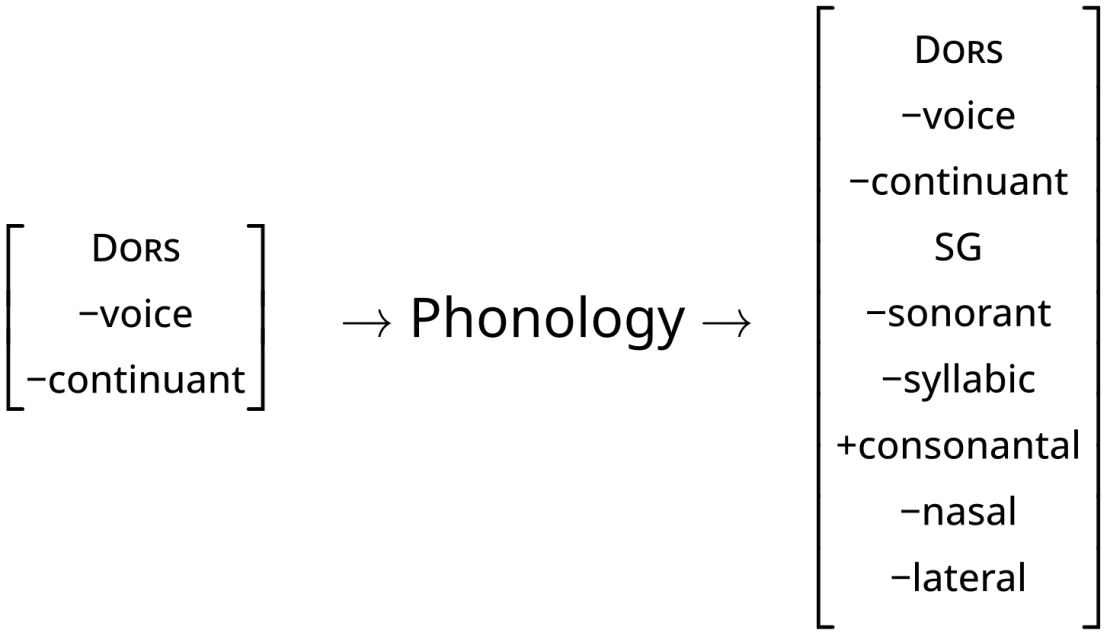

## Distinctive Features

A “fully specified” phone has every possible feature specified

$$k^h$$

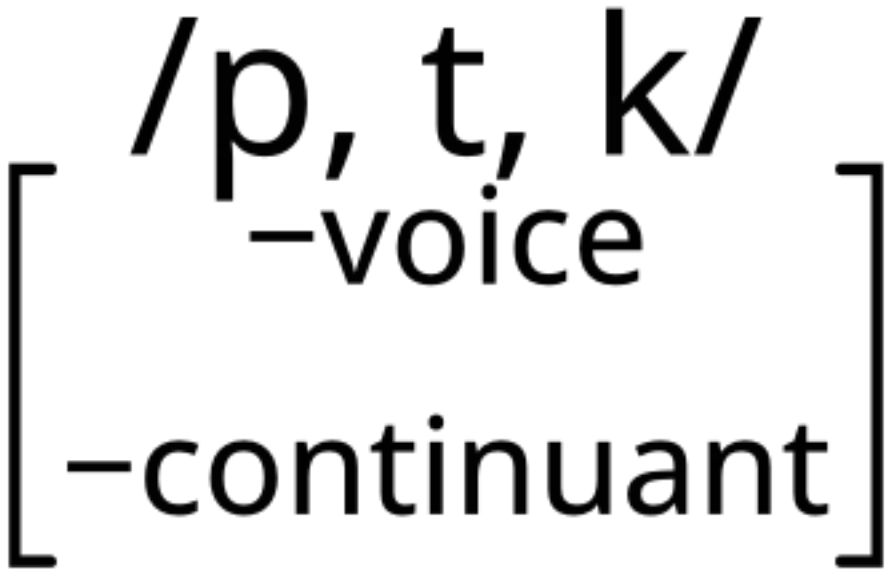
$$\begin{bmatrix} \text{DORS} \\ -\text{voice} \\ -\text{continuant} \\ \text{SG} \\ -\text{sonorant} \\ -\text{syllabic} \\ +\text{consonantal} \\ -\text{nasal} \\ -\text{lateral} \end{bmatrix}$$

## Distinctive Features



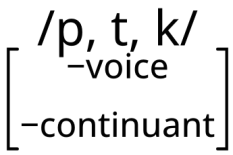

## Distinctive Features

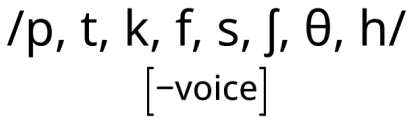
A set of distinctive features describes a natural class




## Distinctive features

The fewer the features, the larger the class



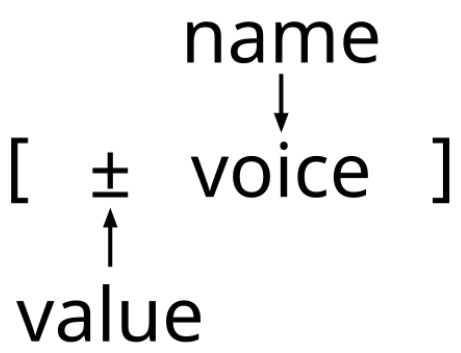




## Distinctive Features - Universal?

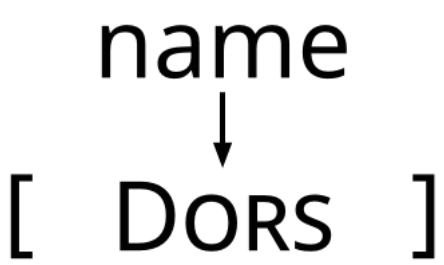
There are conventionalized names for most distinctive features, and some researchers suggest there is a universal set of features that each language can draw from.

## Distinctive Features - Notation

“Binary” features




“Monovalent” features




## Major Class

**[± syllabic]**

[+] = Can be nucleus of a syllable.

**[± consonantal]**

[-] = No significant oral tract constrictions. (Often shortened to “V” and “C”)

## Major Class

| | [-syllabic] | [+syllabic] |
|---|---|---|
| [+consonantal] | Your typical C | [bʌʔn̩, wɔtɻ̩] |
| [-consonantal] | [jɛs] | Your typical V |

## Manner

**[±sonorant]**

[-] = build up of pressure behind oral constriction

**[±continuant]**

[-] = complete closure

**[±nasal]**

[+] = open velum

**[±lateral]**

[+] = closure down the center, open to the sides

## Manner

| | [+sonorant] | [-sonorant] |
|---|---|---|
| [+continuant] | glides, approximants | fricatives |
| [-continuant] | nasal stops | oral stops |

## Laryngeal Features

**[±voice]**

[+] = vocal fold vibration

**[SpreadGlottis]**

Large glottal opening

**[ConstrictedGlottis]**

Constricted vocal folds

## Laryngeal Features

| | | [SG] |
|---|---|---|
| [+voice] | [b, d, g] | [bʱ, dʱ, gʱ] (breathy) |
| [-voice] | [p, t, k] | [pʰ, tʰ, kʰ] |

## “Laryngeal Realism”

Some people argue that [±voice] is actually [Voice].

**English**

| Features | Segments |
|---|---|
| [] | /b, d, g/ |
| [SG] | /p, t, k/ |

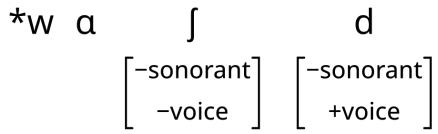
**Dutch**

| Features | Phonemes |
|---|---|
| [Voice] | /b, d, g/ |
| [] | /p, t, k/ |

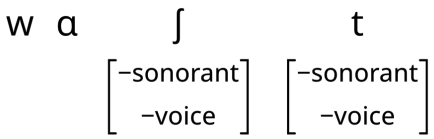
## Underspecification

Not every phoneme that has phonetic voicing is necessarily [+voice]. e.g. anything [+sonorant] might get voicing “for free.”

## Underspecification

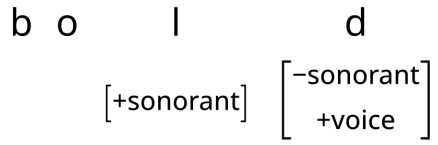



washed

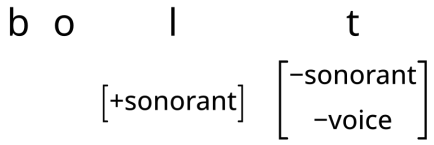



washed

## Underspecification




bold




bolt

## Major Place Features

All privative or monovalent features

**[Labial]**

Involving the lips.

**[Coronal]**[1]

Front part of the tongue

**[Dorsal]**[2]

Back part of the tongue

**[Pharyngeal]**

Involving the pharynx

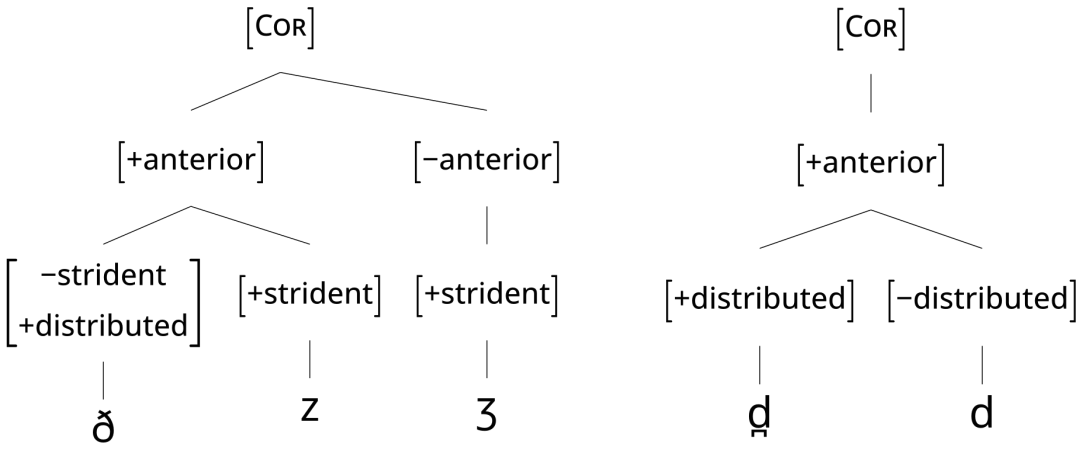
**[Laryngeal]**

Constriction at the glottis

[1]Not to be confused with the “coronal plane” from articulatory imaging!
[2]Not to be confused with the “dorsal plane” from articulatory imaging!

## Subsidary Place Features

The broad strokes place features can't capture all contrastive places differences

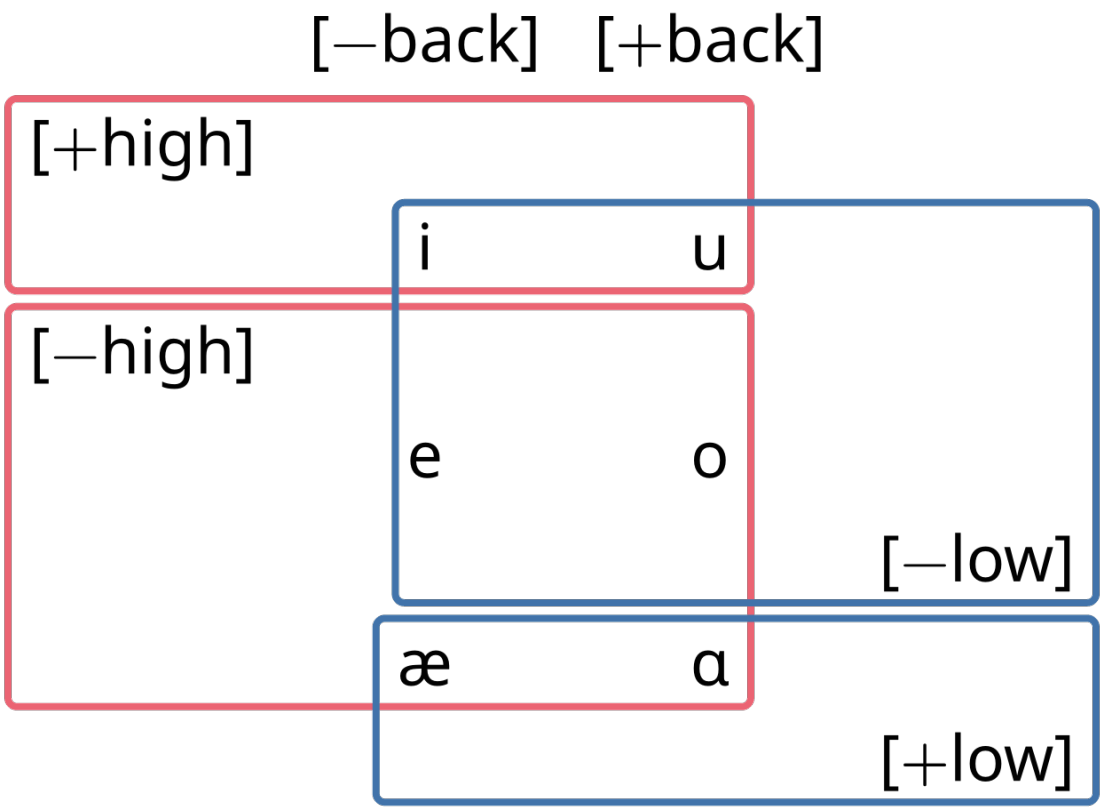



## Vowels

Vowels are all going to be

$$\begin{bmatrix} \text{Dors} \\ +\text{syllabic} \\ -\text{consonantal} \end{bmatrix}$$

## Vowels




## Vowels

**[±round]**

| [-round] | [+round] |
|---|---|
| ʌ | o |
| ɑ | ɔ |

**[±ATR]** or **[±tense]**

| [+ATR] | [-ATR] |
|---|---|
| i | ɪ |
| e | ɛ |
| u | ʊ |

## Feature Theories

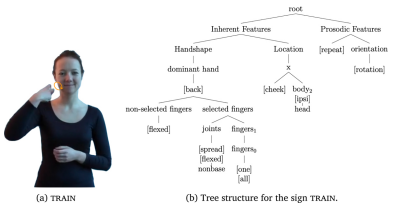
Sometimes, the feature set we have isn't quite adequate.

In Philadelphia, /æ/ gets tensed before the following segments in red:

| | | | |
|---|---|---|---|
| p | | t | k |
| b | | d | g |
| f | θ | s | ʃ |
| v | ð | z | ʒ |
| m | | n | ŋ |

## Feature Theories

Signed languages also have distinctive features




(a) TRAIN (b) Tree structure for the sign TRAIN.

Figure 1: "TRAIN" in French Sign Language (Mertz et al., 2022)

## Using Distinctive Features

Turkish Vowel Harmony

| *translation* | nominative | accusative |
|---|---|---|
| *wind* | jel | jel-i |
| *tooth* | diʃ | diʃ-i |
| *rose* | gyl | gyl-y |
| *sea* | gøl | gøl-y |
| *girl* | kɯz | kɯz-ɯ |
| *branch* | dɑl | dɑl-ɯ |
| *arm* | kol | kol-u |

## Using Distinctive Features

Turkish Vowel Harmony

- Accusative morpheme: [+high]

- Assimilates to the preceding vowel for [±back] and [±round]

## References

## Bibliography

Mertz, J., Annucci, C., Aristodemo, V., Giustolisi, B., Gras, D., Turco, G., Geraci, C., & Donati, C. (2022). Measuring sign complexity: Comparing a model-driven and an error-driven approach. *Laboratory Phonology*, *24*(1). https://doi.org/10.16995/labphon.6439